

Unemployment Analysis with Python

1. Introduction

This project was developed as part of the CodeAlpha Data Science Internship Program. The purpose of this project is to analyze unemployment trends in India using Python and data visualization techniques.

Unemployment is a key economic indicator that reflects the health of a country's economy. Through this analysis, patterns and trends in unemployment data can be identified to support informed decision-making.

2. Objective

The main objectives of this project are:

- To analyze unemployment rate data in India.
 - To perform data cleaning and preprocessing.
 - To visualize unemployment trends.
 - To investigate the impact of COVID-19 on unemployment rates.
 - To identify regional and temporal unemployment patterns.
 - To generate meaningful insights from the data.
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3. Dataset Description

The dataset used for this project is:

Unemployment in India.csv

Features Included

- Region
- Date
- Frequency
- Estimated Unemployment Rate (%)
- Estimated Employed
- Estimated Labour Participation Rate (%)
- Area

Dataset Summary

- Original Records: 768
- Records after Cleaning: 740

- Missing Values Removed: 28

The dataset contains unemployment statistics for different regions across India and includes both urban and rural areas.

4. Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
 - Removed leading and trailing spaces from column names.
 - Handled missing values.
 - Converted the Date column into datetime format.
 - Verified data structure and consistency.
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5. Exploratory Data Analysis

Several analytical techniques were used to explore the dataset.

Analyses Performed

- Distribution of Unemployment Rate
- Top States by Average Unemployment Rate
- Urban vs Rural Unemployment Comparison
- Monthly Trend Analysis
- COVID-19 Impact Analysis
- Correlation Heatmap

These visualizations helped uncover trends and relationships within the dataset.

6. COVID-19 Impact Analysis

A dedicated analysis was performed to study unemployment trends during the COVID-19 period.

Observations indicated a significant rise in unemployment rates after the onset of the pandemic, highlighting the economic disruption caused by lockdowns and reduced business activity.

7. Key Insights

- Unemployment rates varied significantly across regions.
 - Certain states consistently recorded higher unemployment levels.
 - Urban and rural unemployment patterns differed.
 - COVID-19 had a substantial impact on employment conditions.
 - Labour participation rates showed relationships with unemployment trends.
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8. Conclusion

The project successfully demonstrated how data science techniques can be applied to analyze unemployment trends and economic indicators.

Through data cleaning, visualization, and trend analysis, valuable insights were generated regarding regional unemployment patterns and the impact of COVID-19 on employment in India.

This project highlights the importance of data-driven decision-making in addressing socio-economic challenges.

9. Tools and Libraries Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
 - Visual Studio Code
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